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The role of GABA in semantic memory and its neuroplasticity

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Abstract

A fundamental aspect of neuroscience is understanding neural functioning and plasticity of the brain. The anterior temporal lobe (ATL) is a hub for semantic memory, which generates coherent semantic representations about the world. GABAergic inhibition plays a crucial role in shaping human cognition and plasticity, but it is unclear how this inhibition relates to human semantic memory. Here, we employed a combination of continuous theta burst stimulation (cTBS), MR spectroscopy and fMRI to investigate the role of GABA in semantic memory and its neuroplasticity. Our results demonstrated that the inhibitory cTBS increased regional GABA levels in the ATL and decreased ATL blood-oxygen level-dependent (BOLD) activity during semantic processing. Importantly, changes in GABA levels were strongly associated with changes in regional activity induced by cTBS. These results suggest that GABAergic activity may be the mechanism by which cTBS induces after effects on cortical excitability. Furthermore, individuals with better semantic performance exhibited selective activity in the ATL, attributable to higher concentrations of inhibitory GABA, which can sharpen distributed semantic representations, leading to more precise semantic processing. Our results revealed a non-linear, inverted-U-shape relationship between GABA levels in the ATL and semantic performance, thus offering an explanation for the individual differences in the cTBS effect on task performance. These results provide neurochemical and anatomical specificity in shaping task-related cortical activity and behaviour. Understanding the link between neurochemistry and semantic memory has important implications for understanding individual differences in semantic behaviour and developing therapeutic interventions for patients with semantic impairments.

eLife assessment

This is a **valuable** paper that might contribute new insight into the role of GABA in semantic memory, which is a significant question in higher cognition. However, the empirical support for the main claims is **incomplete**, with some results not fully coherent and robust – the paper would benefit from more rigorous analyses. These results, once strengthened, will be of interest to broad readers of the neuroscience and cognitive neuroscience community.

Main

Understanding how the brain functions to drive flexible human behaviour has been a fundamental challenge in cognitive neuroscience. The ability to (re)shape our behaviours based on our experiences relies on a flexible mechanism in the brain, which is achieved through the regulation of neural excitation and inhibition[1]. In particular, an imbalance between excitatory and inhibitory processes has been associated with various cognitive impairments in several psychiatric disorders[2] such as autism spectrum disorder[3] and schizophrenia[4]. Of particular interest is the role of the neurotransmitters, gamma-aminobutyric acid (GABA) and glutamate in coordinating neural functions supporting performance in various cognitive domains. GABA is the primary inhibitory neurotransmitter in the brain, while glutamate is the primary excitatory neurotransmitter. The balance between GABAergic and glutamatergic neurotransmission is crucial for the proper functioning of the brain and the maintenance of optimal behavioural responses in both healthy and diseased states[5–7]. While GABA and glutamate are associated with various cognitive processing, the mechanistic link from the neurotransmitters to human cognition is not well understood.

Research has shown that GABAergic inhibition plays a crucial role in various processes, such as sensory processing, attention, memory and learning[5,6,8]. Furthermore, GABAergic neurotransmission can regulate synaptic plasticity, leading to long-term potentiation (LTP) and long-term depression (LTD) by modulating the activity of excitatory neurons[9–11]. Understanding GABAergic inhibition is crucial in uncovering the neurochemical mechanisms underlying human cognition and its neuroplasticity. However, the relationship between GABAergic inhibition and higher cognition in humans remains unclear. Additionally, variability in the levels of GABA in the human brain may contribute to individual differences in cognitive behaviour[12,13]. The aim of the study was to investigate the role of GABA in relation to human higher cognition – semantic memory and its neuroplasticity at individual level.

Semantic memory is a crucial aspect of human cognition, encompassing our knowledge of concepts and meaning, such as words, people, objects and emotion[14,15]. Accumulating and converging evidence indicates that the anterior temporal lobe (ATL) is a transmodal and transtemporal hub of semantic memory that generates coherent semantic representations though interactions with modality-specific brain regions and integration over time/episodes[15–17]. The initial and strong evidence supporting this hypothesis comes from semantic dementia patients who show selective semantic degradation in both verbal and non-verbal domains due to progressive ATL-centred atrophy[14,18–20]. Recent studies also have supported this hypothesis using intracranial recordings and cortical stimulation[21,22], magnetoencephalography[23,24] and functional magnetic resonance imaging (fMRI)[25–28]. Transcranial magnetic stimulation (TMS) studies have further established the link between ATL and semantic memory. Perturbing the ATL with inhibitory repetitive TMS (rTMS) and theta burst stimulation (TBS) made healthy individuals mimic semantic dementia patients, showing slower reaction time on semantic processing[29–35]. Our investigations combining rTMS/TBS with fMRI have revealed the critical role of the ATL in the neuroplasticity of the semantic system, demonstrating the flexible and adaptive nature of the neural mechanisms underpinning semantic memory function[29,36,37]. Despite the compelling and consistent findings regarding the involvement of the ATL in semantic memory and its capacity for neuroplasticity, the specific ways in which the underlying neurotransmitter systems influence ATL function in semantic memory and its neuroplasticity remain unclear.

Previously, we explored neurotransmitter systems on the functioning of the anterior temporal lobe (ATL) in semantic memory using a combination of magnetic resonance spectroscopy (MRS) and fMRI[38]. By utilizing MRS, a non-invasive method for measuring neurometabolites such as GABA and glutamate *in vivo*, we were able to detect and quantify regional GABA and glutamate concentrations in the ATL[39]. The concentration of GABA in the ATL showed a positive correlation with performance in semantic tasks and was negatively associated with blood-oxygen level-dependent (BOLD) signal changes during semantic processing. Our results highlighted the critical involvement of GABAergic inhibition in the modulation of neural activity and behaviour related to semantic processing in the ATL. Subsequently, we explored the relationship between regional GABA levels in the ATL and cTBS-induced plasticity in semantic memory[40]. To achieve this, we acquired the ATL MRS and fMRI prior to stimulation and then delivered cTBS, an inhibitory protocol[41], to the ATL. We examined how baseline GABA levels in the ATL were associated with changes in semantic task performance after cTBS. The results showed that individuals with higher GABA levels in the ATL exhibited stronger cTBS effects on semantic processing, especially those who displayed inhibitory responses after cTBS. These findings suggest that the GABAergic action in the ATL plays a crucial role in cTBS-induced plasticity in semantic memory, predicting inter-individual variability of cTBS responsiveness.

Based on these findings, we hypothesized that GABAergic inhibition in the ATL may affect neural dynamics of the ATL underpinning semantic memory and its neuroplasticity. We used a combination of cTBS, MRS and fMRI to examine the relationship between changes in GABA levels, cortical activity during a semantic task, and semantic task performance. First, we hypothesized that the inhibitory cTBS would increase GABA concentrations and decrease task-induced BOLD signal changes in the ATL. Second, the effects of cTBS on semantic processing could be attributed to GABAergic action in the ATL at the individual level: greater changes in GABA concentrations would result in increasing changes in task-induced BOLD signal during semantic processing and semantic task performance. Additionally, to address and confirm the relationship between regional GABA levels in the ATL and semantic memory function, we combined data from our previous study[38] with the current study's data. We then explored the function of GABA in the ATL in relation to semantic function. Finally, we extended this GABAergic function to semantic neuroplasticity driven by cTBS.

Results

We acquired resting-state MRS for the ATL and vertex followed by fMRI before and after cTBS (**Fig. 1A**). During fMRI, participants made semantic association decisions as an active task and pattern matching as a control task (**Fig. 1B**). GABA concentrations were estimated from the ATL with the vertex as a control region (**Fig. 1C**). cTBS with 80% of resting motor threshold (RMT) was delivered outside of scanner at one of the target regions with a week gap between two sessions (**Fig. 1D**).

cTBS modulates regional GABA concentrations and task-related BOLD signal changes in the ATL

E-field modelling of cTBS showed that, as intended, ATL cTBS stimulated the left ventrolateral ATL (**Fig. 2A**). To investigate how cTBS modulates GABA concentrations, we quantified GABA/NAA and calculated the changes (POST– PRE). A 2 × 2 repeated measures analysis of variance (ANOVA) with stimulation (ATL vs. vertex) and VOI (ATL vs. vertex) as within subject factor was performed. There was a significant interaction effect between the stimulation and VOI ($F_{1,16}$ = 4.57, p = 0.048) (**Fig. 2B**). There was no significant main effect of the stimulation ($F_{1,16}$ = 3.23, p = 0.091) and VOI ($F_{1,16}$ = 0.64, p = 0.435). Planned paired t-tests revealed that ATL stimulation significantly increased GABA concentrations in the ATL compared to the control stimulation (t = 1.86, p = 0.040) and control site (t = 2.07, p = 0.027). There were no cTBS effects in the vertex VOI regardless of the stimulation (ps > 0.23). It is noted that there was no significant cTBS effect in Glx (*Fig. S1*).

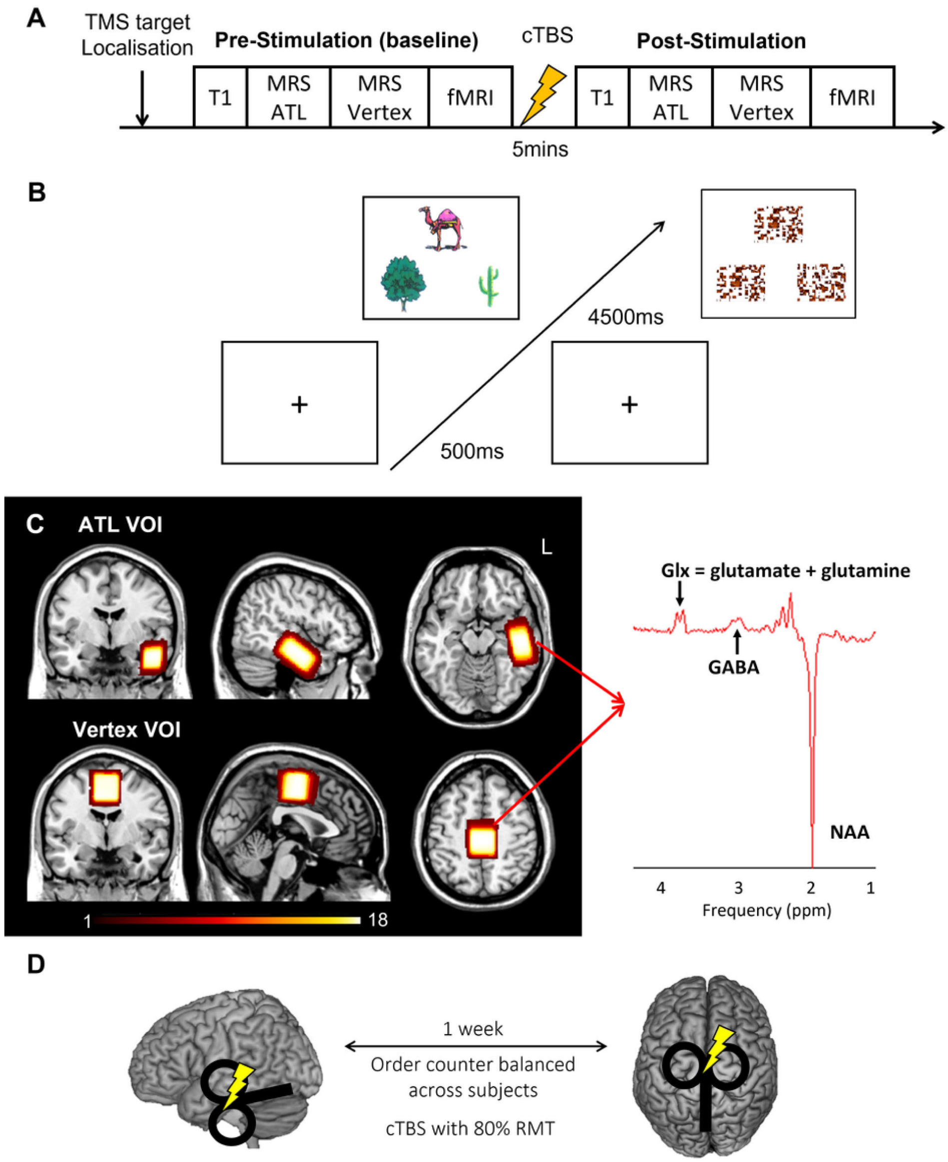

Figure 1

A) Experimental procedure. B) An example of the semantic association task (left) and control task (right: pattern matching). Each trial starts with a fixation followed by stimuli, which have 3 items, a target on the top and 2 choices at the bottom. C) The location of volume of interest (VOI) for MRS (left: ATL and vertex) and a representative MRS spectrum with estimated peaks (right). Colour bar indicates the number of overlapping participants. NAA: N-acetylaspartate. D) cTBS protocols. cTBS was applied over the left ATL and vertex as a control site. Each stimulation was delivered on different days with a week gap at least.

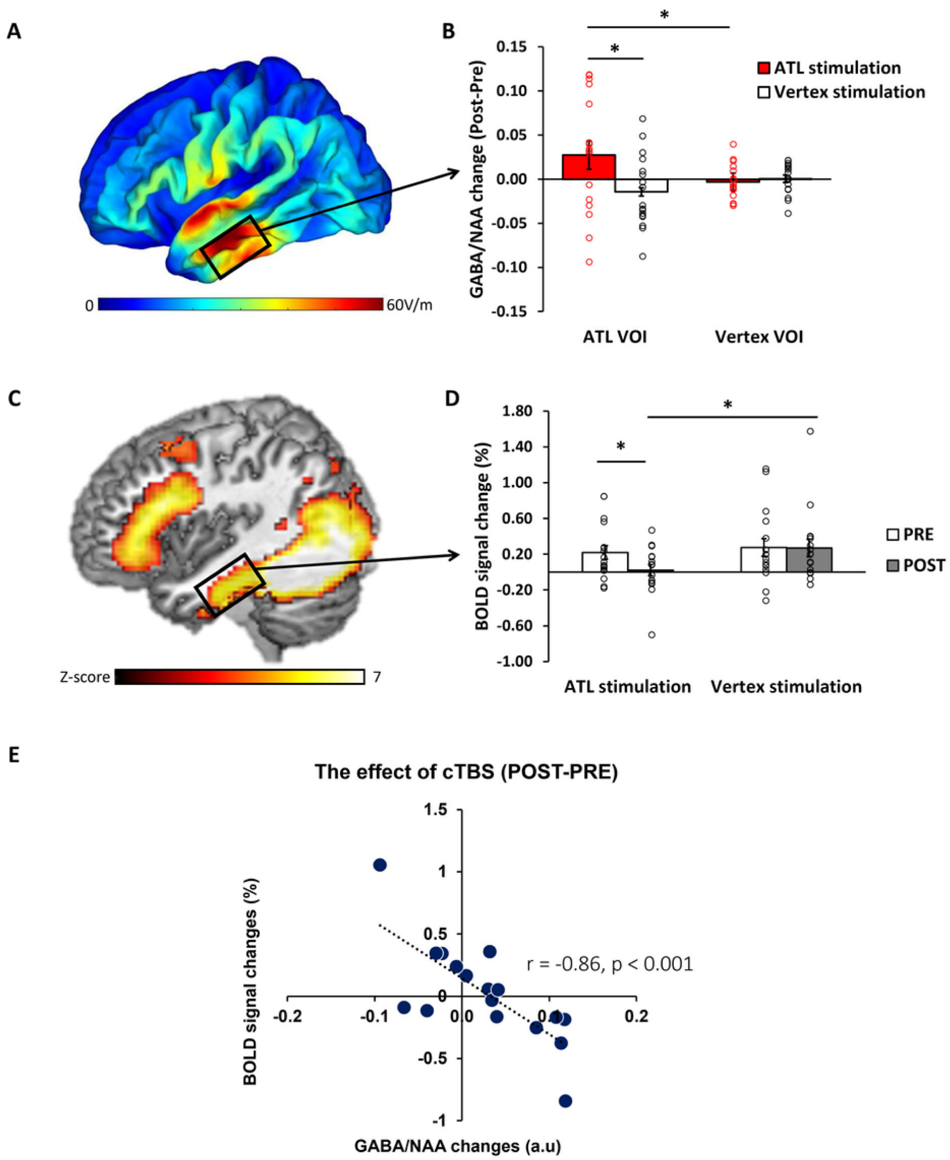

Figure 2

The effects of cTBS in the ATL. A) ATL cTBS e-field modelling. B) cTBS-induced regional GABA changes in the ATL. Red bar indicates the ATL stimulation and white bar indicates the control (vertex) stimulation. C) fMRI result of the contrast of interest (semantic > control). D) cTBS-induced ATL BOLD signal changes. White bars represent the pre-stimulation session, and grey bars represent the post-stimulation session. E) The relationship between cTBS-induced GABA changes and BOLD signal changes in the ATL. Each individual is represented as a circle. * $p < 0.05$

fMRI results demonstrated that the semantic association task evoked increased activation in the ATL, prefrontal and posterior temporal cortex compared to the control task (**Fig. 2C**). To examine the effects of cTBS, we performed ROI analysis using the same VOI in the ATL. Planned paired t-tests revealed that BOLD signal changes during semantic processing were significantly altered after ATL cTBS compared to the pre-stimulation ($t = 1.78$, $p = 0.046$) and the control stimulation ($t = -2.11$, $p = 0.025$) (**Fig. 2D**).

To investigate the effects of ATL cTBS, we conducted a partial correlation analysis between GABA changes (POST–PRE) and BOLD signal changes (POST– PRE), accounting for age and sex. We found a significant correlation between cTBS induced GABA changes and BOLD signal changes in the ATL ($r = -0.86$, $p < 0.001$). Individuals with greater increases in ATL GABA levels following ATL cTBS showed greater reductions in task-induced BOLD signal changes in the ATL. These results demonstrate that ATL cTBS modifies regional GABA concentrations, and the cTBS-induced changes in GABA levels are connected to individual-level changes in task-related fMRI signal.

Participants' performance was examined using a 2 × 2 repeated measures ANOVA with stimulation (ATL vs. vertex) and session (PRE vs. POST) as within-subject factors. There were no significant main effect and interaction on reaction time (RT) in the semantic task (Fs > 0.19, ps > 0.220). However, we found a significant main effect of session in the control task ($F_{1, 15} = 20.21$, $p < 0.001$). *Post hoc* paired t-tests demonstrated that participants performed the task faster in the post-session compared to the pre-session, except in the semantic task after the ATL stimulation (*Table 1*). The results showed that ATL cTBS attenuated the practice effects found in the control stimulation and control task. There were no significant effects in accuracy (Fs > 0.01, ps > 0.073).

Regional GABA concentrations in the ATL play a crucial role in semantic memory

In our prior study[38], ATL GABA levels were significantly and negatively correlated with ATL activity during semantic processing (**Fig. 3A**). Here, we replicated our previous findings in a different cohort with the same research paradigm (pre-stimulation session). We conducted a single-voxel regression analysis with the individual's GABA concentrations (ATL pre-stimulation session) as the regressor of the fMRI contrast of interest (semantic > control). The BOLD response in the ventral ATL was significantly and negatively correlated with the individual GABA levels in the ATL (MNI − 42 − 6 -33, $p_{\text{SVC−FWE}} < 0.05$), overlapping with the results from our previous study[38] (**Fig. 3A**). The GABA-related region of the ventral ATL overlapped with the semantic coding hotspot from electrocorticograms (ECoG) data and direct cortical stimulation[21,22] (**Fig. 3A**). Furthermore, we found that individual GABA concentrations in the ATL were positively associated with semantic task performance (**Fig. 3B**). We also confirmed this finding, demonstrating that individuals with more GABA in the ATL performed the semantic task better (higher accuracy) ($r = 0.50$, $p = 0.035$) (**Fig. 3B**). It should be noted that individual GABA levels also significantly correlated with ATL activity and semantic task performance at the vertex stimulation session (*Fig. S2*). These results demonstrate that higher levels of cortical GABA in the ATL are associated with task-related regional activity as well as enhanced semantic function.

The inverted U-shaped function of ATL GABA concentrations in semantic processing

The pattern of correlation between GABAergic activity and semantic task accuracy observed in our previous study was replicated in an entirely new cohort in the current study. Next, we combined the two studies (N = 37) in order to fully investigate the potential role of GABA in the ATL as a mechanistic link between ATL inhibitory GABAergic action and semantic task performance. First, we tested the linear relationship between ATL GABA levels and semantic task performance. We confirmed our previous findings that individuals with higher GABA levels in the

		Accuracy (%)				Reaction time (s)				
		PRE		POST		PRE		POST		
		mean	SD	mean	SD	mean	SD	mean	SD	
ATL stimulation	Semantic	82	8	82	6	1.99	0.44	1.97	0.44	t = 0.37, p = 0.720
	Control	97	5	98	2	1.71	0.24	1.53	0.26	t = 3.27, p = 0.005
Vertex stimulation	Semantic	78	12	80	10	2.11	0.40	2.00	0.43	t = 2.65, p = 0.017
	Control	95	5	96	5	1.75	0.36	1.64	0.36	t = 2.24, p = 0.040

Table 1

The results of behavioural performance. SD represent the standard deviation.

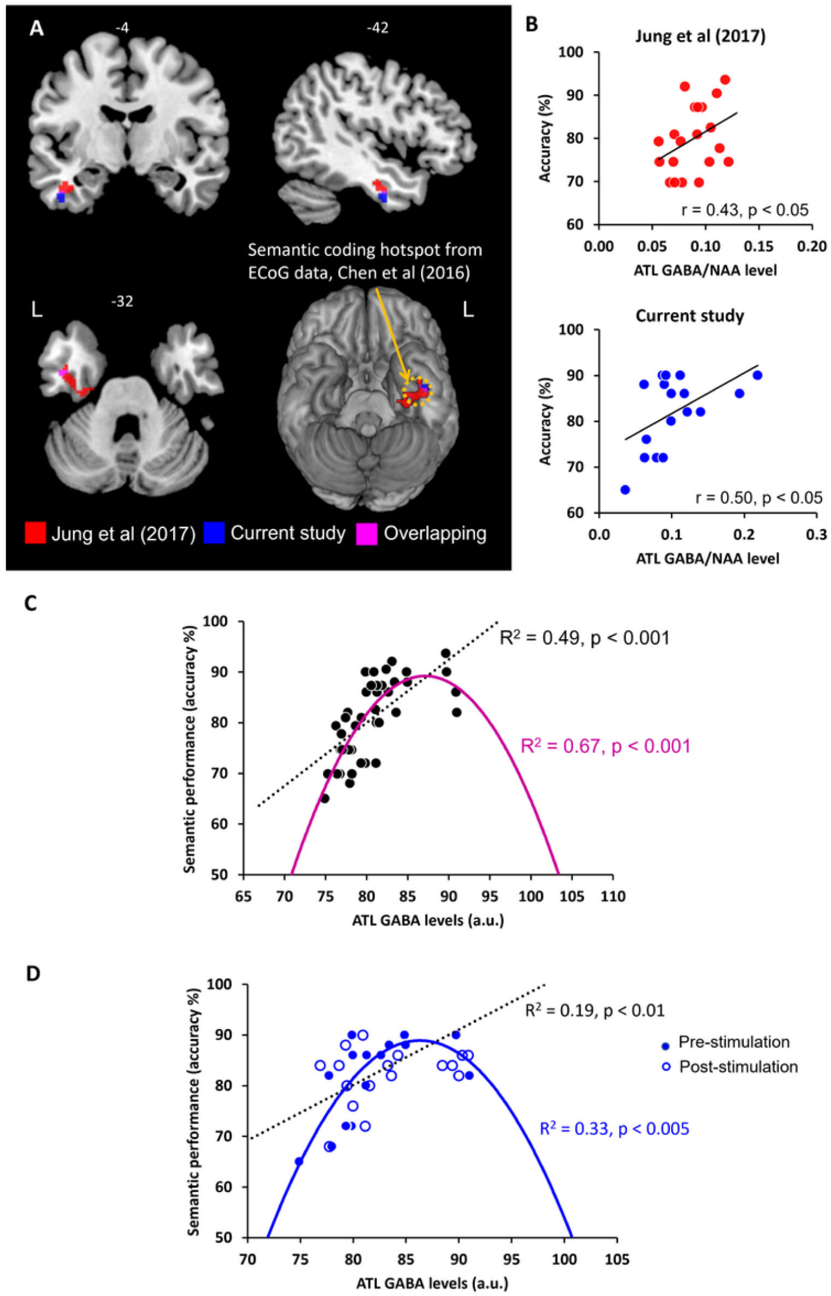

Figure 3

A) Local maxima of the voxel-wise regression analysis of the contrast (semantic > control) with GABA concentrations in the ATL. B) The relationship between individual GABA levels in the ATL and semantic task performance from our previous study (Jung et al., 2017) and current study (pre-stimulation session). C) The ATL GABA function in relation to semantic performance. D) The relationship between cTBS-induced changes in ATL GABA levels and semantic task performance. Dotted line represents the linear function between ATL GABA levels and semantic task performance. Coloured line represents the inverted U-shaped (quadratic) function between ATL GABA levels and semantic task performance.

ATL showed better semantic task performance (R^2 = 0.49, p < 0.001) (**Fig. 3C**). Second, to test our hypothesis, we assumed that semantic performance follows an inverted U-shaped (quadratic) function with relation to ATL GABA concentrations. In other words, people who have low or excessive GABA levels in the ATL perform the semantic task relatively poorly. The results revealed that the inverted U-shaped function between ATL GABA and semantic performance was significant (R^2 = 0.67, p < 0.001) (**Fig. 3C**). To compare two different models, we calculated the Bayesian Information Criterion (BIC) as a measure of model fitness[42] and performed a partial F-test to determine whether there is a statistically significant difference between two models. A best model fitness can be characterized by low BIC and high R^2. The results showed a BIC value of 243.72 for the linear function and a value of 233.36 for the quadratic function. The results of F-tests revealed that the inverted U-shaped model provided a statistically significantly better fit than the linear model (F = 15.60, p < 0.001). The best-fitting model is therefore the inverted-U-shaped function of ATL GABA in semantic processing.

We performed the same analysis on the pre- and post-stimulation data in order to investigate the role of ATL GABA in semantic plasticity. We found that there was a significant linear relationship between the ATL GABA levels and semantic performance before and after stimulation (R^2 = 0.19, p < 0.01) (**Fig. 3D**). The inverted U-shaped function also showed a significant association between them (R^2 = 0.33, p < 0.005) (**Fig. 3D**). The F-test demonstrated that the quadratic model showed a significantly better fit than the linear model (F = 6.64, p = 0.014). The inverted U-shaped function has the better BIC score for explaining changes in ATL GABA levels and semantic performance induced by cTBS (linear model BIC 230.21, quadratic model BIC 227.13). Thus, the best-fitting model is the inverted U-shaped for the ATL GABA changes induced by cTBS in relation to semantic function.

Discussion

We investigated the role of cortical GABA in the ATL on semantic memory and its neuroplasticity. Our results provide strong evidence that regional GABA levels increase following inhibitory cTBS in human associative cortex, specifically in the ATL, a representational semantic hub. Notably, the observed increase was specific to the ATL and semantic processing, as it was not observed in the control region (vertex) and not associated with control processing (visuospatial processing). Our study also found that the magnitude of cTBS-modulated GABA changes at the individual level was associated with their changes in ATL activity during semantic processing. Furthermore, our data confirmed and replicated our previous findings that GABA concentrations in the ATL shape task-related cortical activity and semantic task performance. In other words, individuals with greater semantic performance exhibit selective activity in the ATL due to higher concentrations of inhibitory GABA. GABAergic inhibition can sharpen activated distributed semantic representations through lateral inhibition, leading to improved semantic acuity[38], which aligns with theories on representational sharpening in visual perception[43,44]. Importantly, our data revealed, for the first time, a non-linear, inverted-U-shape relationship between GABA levels in the ATL and semantic function, by explaining individual differences in semantic task performance and cTBS responsiveness. Understanding the link between neurochemistry and semantic memory is an important step in understanding individual differences in semantic behaviour and could guide therapeutic interventions to restore semantic abilities in clinical settings.

To the best of our knowledge, this is the first study to demonstrate that (1) cTBS modulates both regional GABA concentrations and cortical activity in human higher cognition - semantic memory, and that (2) changes in GABA levels are closely linked to changes in regional activity induced by cTBS. These results suggests that GABAergic activity may be the mechanism by which cTBS induces long-lasting after-effects on cortical excitability, leading to behavioural changes. Previous studies in animals and humans have also suggested that cTBS can induce LTD-like effects on cortical synapses and is associated with the GABAergic system in the cortex[45–50]. Another study employing MRS found that cTBS increased regional GABA concentrations at the primary motor cortex in healthy subjects[51]. These findings suggest that cTBS activates a population of cortical GABAergic interneurons, leading to the increase in GABAergic activity[52,53]. As a major inhibitory neurotransmitter, GABA has been shown to have a negative correlation with BOLD signal changes[5,54]. Previous, we demonstrated this negative relationship between ATL GABA levels and BOLD signal changes in the ATL during semantic processing[38], indicating a potential role of GABA in shaping the functions/computations of the cortex. Here, we further demonstrated that increase in GABA induced by cTBS was negatively correlated with the reduction of BOLD signal responses in the ATL following cTBS, during semantic processing. Our findings suggest a crucial role for GABAergic inhibition in the ATL shaping the local neural functioning underpinning semantic memory and its neuroplasticity. The GABAergic inhibition confines the propagation of excitatory signalling, thereby maintaining the functional organization of the cortex[55], and the modulation of cortical GABAergic inhibition drives experience-dependent plasticity in cognition[10,56].

GABA exists in two distinct neuronal pools: cytoplasmic GABA, which is involved in metabolism, and vesicular GABA, which plays a role in inhibitory synaptic neurotransmission[57]. In addition to intracellular GABA, extracelluar GABA exerts tonic inhibition through extra-synaptic $GABA_A$ receptors[58]. MRS is capable of detecting the total concentration of GABA in the voxel of interest, but it cannot differentiate between different pools of GABA[59,60]. Some studies have suggested that MRS-measured GABA signals reflect GABAergic tonic inhibition rather than synaptic GABA signalling[61,62] whereas other studies have failed to replicate this relationship[63,64]. A recent study has shown a link between MRS-measured GABA and phasic synaptic GABAergic activity[65]. Although findings of previous studies have been mixed, changes in GABA levels observed in this study may reflect cTBS-modulated GABAergic neurotransmission, which encompass both tonic and synaptic GABAergic activity. This GABAergic activity shapes the selective response profiles of neurons in the cortex[9].

Inverted U-shaped models have been previously considered in the field of neuroscience, specifically in terms of the relationship between the concentration of neurotransmitters such as dopamine, acetylcholine and noradrenaline, and the level of neural activity[66–69]. Recent studies suggest that this relationship also applies to behaviour, where moderate levels of neural activity are linked to the optimal performance (for a reveiw, see[70]). For example, Ferri et al.[71] showed an inverted U-shaped relationship between excitation and inhibition balance and multisensory integration, where extreme values impair functionality while intermediate values enhance it, even in healthy individuals. Our findings revealed a non-linear relationship between GABA levels in the anterior temporal lobe and semantic function, indicating that individual variations in semantic task performance can be explained by an inverted-U-shape pattern (**Fig. 4A**). Specifically, for relatively greater levels of GABA in the ATL, with lower task-induced regional activity, were associated with better semantic processing in healthy participants[38]. That is, individuals with better semantic memory abilities show more specific cortical activity in the ATL, which is linked to higher concentrations of inhibitory GABA. Extreme levels of GABA can be found in studies with dementia patients and pharmacological studies with GABA agonists. Recent studies have reported decreased GABA levels in Alzheimer's disease[72,73] and frontotemporal dementia[74,75] in relation to their cognitive impairments such as memory and language. In fact, GABA agonists like midazolam have been found to improve a verbal generation in anxiety patients by increasing GABAergic function[76]. On the other hand, healthy participants who received GABA supplementation (such as baclofen) have been found to have decreased task performance[77]. Overall, optimal, elevated levels of GABA in the ATL may aid in refining stimulated widespread semantic representations through local inhibitory processes.

This inverted U-shaped model could also explain inter-individual variability in cTBS-induced neuroplasticity in the ATL in semantic processing. Our data demonstrated that cTBS over ATL increased regional GABA concentrations, but there was inter-individual variability in GABA level

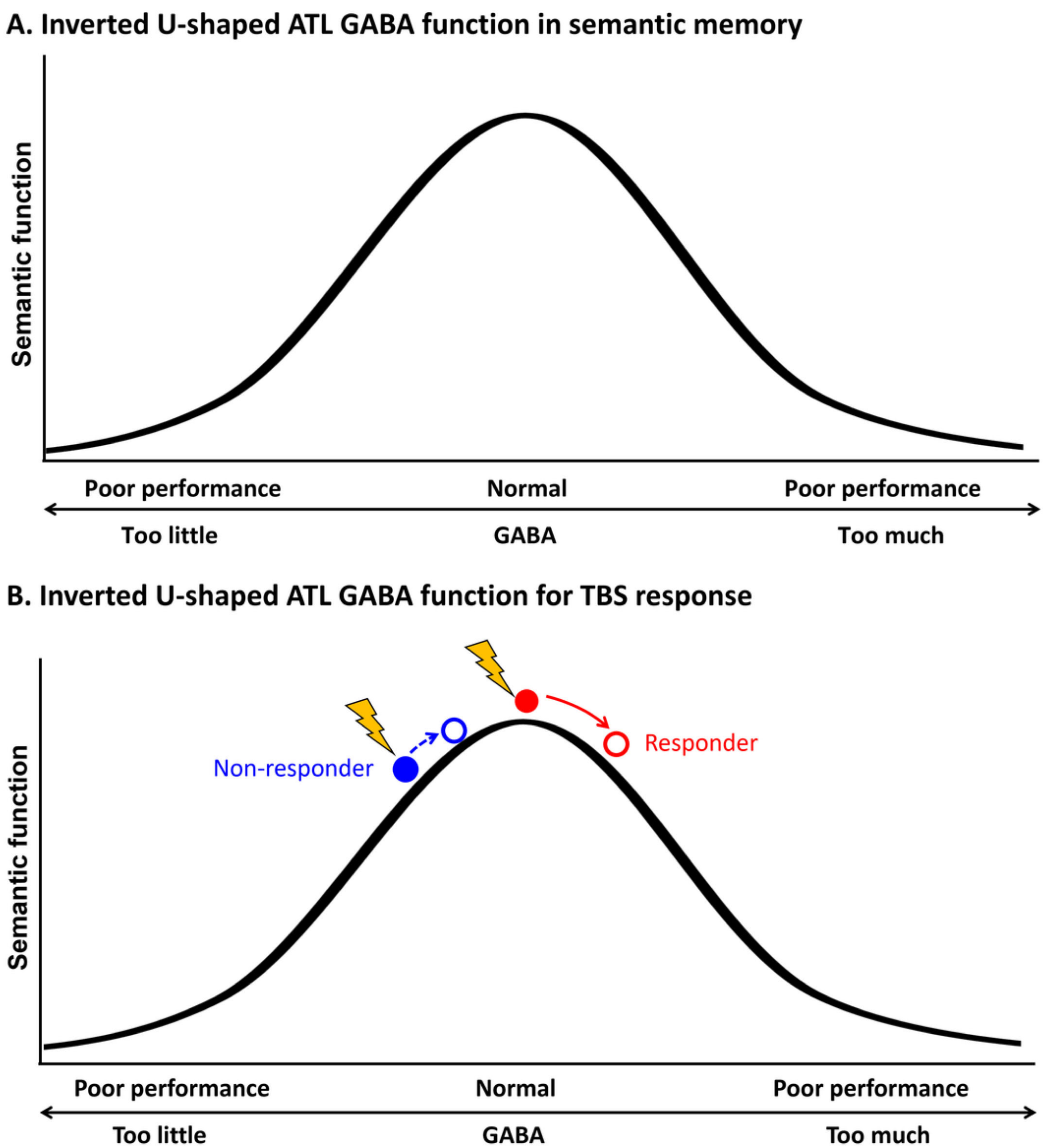

Figure 4

Schematic diagram of relationship between the concentration of GABA in the ATL and semantic function. A) Inverted U-shaped ATL GABA function in semantic memory B) Inverted U-shaped ATL GABA function for cTBS response on semantic memory.

changes in response to cTBS (**Fig. 2**). Our previous investigation[40] showed that the pre-interventional neurochemical state was crucial in predicting cTBS-induced changes in semantic memory. Specifically, cTBS over the ATL inhibited the semantic task performance (i.e., reduced accuracy) of individuals with initially higher concentration of GABA in the ATL, linked to better semantic capacity. However, cTBS had a null or even facilitatory effect on individuals with lower semantic ability with relatively lower GABA levels in the ATL. This study suggests that individuals with higher GABA levels in the ATL were more likely to respond to cTBS, exhibiting inhibitory effects on semantic task performance (responders), while individuals with lower GABA concentrations and lower semantic ability were less likely to respond or even showed facilitatory effects after ATL cTBS (non-responders). The current study revealed a non-linear, inverted-U-shape relationship between GABA levels in the ATL and semantic function, by explaining individual differences in semantic task performance and cTBS responsiveness. As regional GABA increases after cTBS, responders with the optimal level of GABA in the ATL would show poorer semantic performance, whereas non-responders could exhibit no changes or even better semantic performance with GABA increase (**Fig. 4B**). This relationship is similar to the inverted U-shaped relationship between dopamine action in the prefrontal cortex (PFC) and cognitive control, whereby moderate levels of dopamine lead to optimal cognitive performance[78]. The effects of dopaminergic drug on PFC function also depend on baseline levels of working memory performance (for a review, see[67]), explaining the effects of dopaminergic drugs on cognitive performance in individuals with varying working memory capacities[79–81].

Although we expected changes in task performance during semantic processing following cTBS, we only found relatively weak inhibitory effects in semantic task performance – the attenuation of a generalised practice effect. Participants showed practice effects in the second task, except for the semantic task after ATL cTBS. We have demonstrated that sufficient task practice (i.e., 120 trials) was required to detect rTMS-induced behavioural changes in semantic processing[34]. The lack of behavioural changes in response to cTBS may be attributed to the fact that the task practice only involved 20 trials for each task.

Our findings provide novel evidence of a direct link from neurochemical modulations to cortical responses in the brain, highlighting substantial individual variability in semantic memory and plasticity. In addition, the current study represents an important replication and extension of previous findings regarding the role of GABAergic inhibition in semantic memory. These results offer fundamental insights into the mechanisms underlying the maintenance and alteration of functional cortical organization in response to perturbations. Our study has important implications for the development of personalized therapeutic interventions aimed at modulating neurochemical systems to restore or enhance higher cognitive function in humans.

Methods and Materials

Participants

Nineteen healthy English native speakers (9 females, mean age = 25.9 ± 5.8 years, age range: 19–38) participated in this study. The sample size was calculated based on a previous study[29], which indicated that to achieve $\alpha = 0.05$, power = 80% for the critical interaction between TMS and task then $N \geq 17$ were required. A participant completed one session (ATL stimulation) only. All participants were right-handed[82]. All subjects provided informed written consent. The study was approved by the local ethics committee.

To explore the role of GABA in semantic memory function, we used the data previously published[38]. Data from twenty healthy, right-handed native English speakers were included (7 males, mean age = 23 ± 4 years, age range: 20–36).

Experimental design and procedure

Participants were asked to visit two times for the study. In each visit, the target region was identified prior to the baseline scan. Participants had a multimodal imaging (MRS and fMRI). Then, participants were removed from the scanner and cTBS was performed in a separate room. Following cTBS, participants were repositioned into the scanner and had the second multimodal imaging (**Fig. 1A**).

We used the same paradigm for the multimodal imaging from our previous study[38]. During MRS, participants were asked to be relaxed with eyes open. Participants performed a semantic association decision task and pattern matching as a control task during fMRI scanning (**Fig. 1B**). The semantic association decision task required a participant to choose which of two pictures at the bottom of the screen was more related in meaning to a probe picture presented on the top of the screen. The items for the semantic association task were from the Pyramids and Palm Tree test[83] and Camel and Cactus test[18]. Items for the pattern matching task were created by scrambling the pictures used in the semantic association task. In the pattern matching task, a participant was asked to identify which of two patterns at the bottom was visually identical to a probe pattern on the top (**Fig. 1B**). Participants were required to press one of two buttons designating two choices in a trial. In each trial, there was a fixation for 500ms followed by the stimuli for 4500ms. A task block had four trials of each task. There were 9 blocks of each task interleaved (e.g., A-B-A-B) with a fixation for 4000ms during fMRI. Total scanning time was about 8mins. E-prime software (Psychology Software Tools Inc., Pittsburgh, USA) was used to display stimuli and to record responses.

Transcranial magnetic stimulation

A Magstim Super Rapid stimulator (MagStim Company, Whitland, UK) with a figure of eight coil (70mm standard coil) was used to deliver cTBS over the left ATL or vertex with a week gap between the stimulation (**Fig. 1D**). cTBS consisted of bursts containing 3 pulses at 50Hz[41] and was applied at 80% of the resting motor threshold (RMT), previously showed inhibitory effects on semantic processing in the ATL[29]. RMT was established for each individual, defined as the minimum intensity of stimulation required to produce twitches on 5 of 10 trials from the right first dorsal interosseous (FDI) muscle when the participant was at rest. The average stimulation intensity (80% RMT) was 49.2% ranging from 38– 60%.

SimNIBS 3.2[84] was used to calculate individual electric field of cTBS. The pipeline by Nielsen et al[85] was utilized to generate the individual head model consisting of five tissue types: grey matter (GM), white matter (WM), cerebrospinal fluid (CSF), skull, and scalp. Then, the fixed conductivity values implemented in the SimNIBS were applied for each tissue type. The electric field interpolation was performed using Saturnino et al[86] and computed the electrical field at the centre of GM in the ATL. Finally, we averaged the individual electrical field (**Fig. 2C**).

Magnetic resonance imaging acquisition

A 3T Philips Achieva MRI scanner was used to acquire data with a 32-channel head coil with a SENSE factor 2.5. Structural images were acquired using a magnetization prepared rapid acquisition gradient echo (MPRAGE) sequence (TR = 8.4 ms, TE = 3.9 ms, slice thickness 0.9 mm, in-planed resolution 0.94 × 0.94 mm).

MRS data were acquired using GABA-edited MEGA-PRESS sequence[87] (TR = 2000ms, TE = 68ms). The voxel of interest (VOI) was manually placed in the left ventrolateral ATL (voxel size = 40 x 20 x 20mm), avoiding hippocampus or vertex (voxel size = 30 x 30 x 30mm), Cz, guided by international 10–20 electrode system[88] (**Fig. 1.C**). Spectra were acquired in interleaved blocks of four scans with application of the MEGA inversion pulses at 1.95 ppm to edit GABA signal (100 repeats at the

ATL VOI and 75 repeats at the vertex VOI). Measurements from the ATL VOI with current protocol provided a robust measure of GABA and glx concentrations[38,39,89]. A total of 1024 sample points were collected at a spectral width of 2 kHz.

A dual-echo fMRI protocol developed by Halai et al[90] was employed to maximise signal-to-noise (SNR) in the ATL (TR = 2.8 s, TE = 12 ms and 35 ms, 42 slices, 96 × 96 matrix, 240 × 240 × 126 mm FOV, slice thickness 3 mm, in-plane resolution 2.5 × 2.5).

MRS analysis

Java-based magnetic resonance user's interface (jMRUI5.1, EU project *www.jmrui.eu*)[91] was used to analyse MRS data. Raw data were corrected using the unsuppressed water signal from the same VOI, eddy current correction, a zero-order phasing of array coil spectra. Residual water was removed using Hankel-Lanczos singular value decomposition[92]. Advanced Magnetic Resonance (AMARES)[93] was used to quantify neurochemicals including GABA, glx, and NAA. The exclusion criteria for data were as follows: Cramér-Rao bounds > 50%, water linewidths at full width at half maximum (FWHM) > 20 Hz, and SNR < 40. A subject was discarded from the analysis due to poor quality of MRS. GABA and glx values are reported as a ratio to NAA as we previously reported[38].

Statistical Parametric Map (SPM8, *http://www.fil.ion.ucl.ac.uk/spm/*) was used to calculate the contributions of GM and WM to the VOI from the structural image. Then voxel registration was performed using custom-made scripts developed in MATLAB by Dr. Nia Goulden, which can be accessed at *http://biu.bangor.ac.uk/projects.php.en*. The calculation of tissue types within the VOI provided the percentage of each tissue type. As GABA levels are substantially higher (two-fold) in the GM than WM[94], we used GM as a covariate in the analysis. There was no significant difference in GM volume before and after the stimulation ($ps > 0.5$) and a significant correlation between GM volumes before and after stimulation in both VOIs (ATL stimulation: $r = 0.75$, $p < 0.001$ in the ATL, $r = 0.67$, $p = 0.003$ in the vertex; Vertex stimulation: $r = 0.68$, $p = 0.008$ in the ATL, $r = 0.72$, $p < 0.001$). The summary of tissue segmentation is summarised in *Table S1*.

fMRI analysis

fMRI data were processed using SPM8. Dual gradient echo images were realigned, corrected for slice timing, and averaged using in-house MATLAB code developed by Halai et al[90]. The EPI volumes were coregistered into the structural image, spatially normalized to the MNI template using DARTEL(diffeomorphic anatomical registration through an exponentiated lie algebra) toolbox[95], and smoothed with an 8 mm full-width half-maximum Gaussian filter.

A general linear model (GLM) was used to perform statistical analyses. A design matrix was modelled with task conditions, semantic, control, and baseline for each individual along with six motion parameters as regressors. A contrast of interest (sematic > control) for each participant were calculated. One-sample t-test was performed to estimate the contrast of interest at the group-level. Clusters were considered significant when passing a threshold of p FWE-corrected < 0.05, with at least 100 contiguous voxels.

Regions of interest (ROI) analysis was conducted using Marsbar[96]. The mean signal changes of VOIs were extracted for semantic task condition before and after the stimulation.

A voxel-wise simple regression analysis was conducted to identify the local maxima of voxels within the MRS ATL VOI correlating with its BOLD response with GABA levels in the contrast of interest (semantic > control). Local maxima of correlation were estimated on a voxel level, setting the threshold to $p < 0.05$ FWE after small-volume correction.

Statistical analysis

For behavioural data, accuracy and reaction time (RT) were calculated for each individual. A 2 × 2 repeated measures ANOVA with stimulation (ATL vs. vertex) and session (PRE vs. POST) as within-subject factors was performed on each task (semantic and control). *Post hoc* paired t-tests were conducted to examine cTBS effects in pre- and post-stimulation sessions.

Partial correlation analysis was performed to illustrate the relationship between the ATL GABA levels and semantic task performance, accounting for GM volume, age, and sex.

Regression analyses (linear and quadratic models) were conducted to explore the relationship between the ATL GABA levels and semantic task performance. The individual ATL GABA levels were adjusted by GM volume, age, and sex, using multiple regression analysis. In order to determine the best-fit model, we calculated the Bayesian Information Criterion (BIC) as a measure of model fitness[42] and performed a partial F-test.

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Declarations

Conflict of interest

The authors declare no competing financial interests.

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Supplemental Materials

cTBS effects on Glx

We conducted a 2 x 2 x 2 repeated measure of ANVOVA with a stimulation (ATL vs. vertex), session (PRE vs. POST), and VOI (ATL vs. vertex) as within subject factors. The results demonstrated a significant main effect of VOI ($F_{1, 16} = 41.56$, $p < 0.001$) (**Fig. 1**). The other effects did not reach a significant level.

	ATL VOI		Vertex VOI	
	PRE	POST	PRE	POST
ATL stimulation	0.50 ± 0.03	0.47 ± 0.03	0.24 ± 0.03	0.22 ± 0.03
Vertex stimulation	0.47 ± 0.03	0.49 ± 0.02	0.29 ± 0.03	0.27 ± 0.04

Table S1

The summary of tissue type within MRS VOIs. Mean ± SE

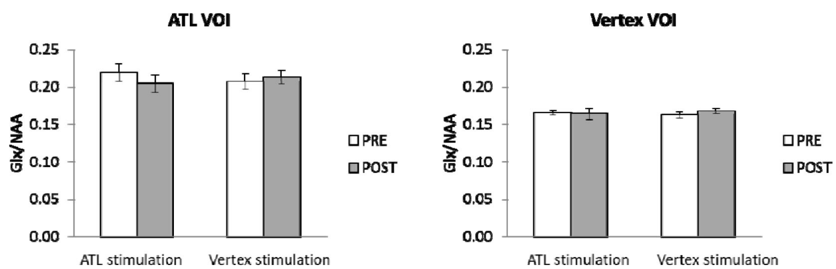

Figure S1.

The effects of cTBS in Glx.

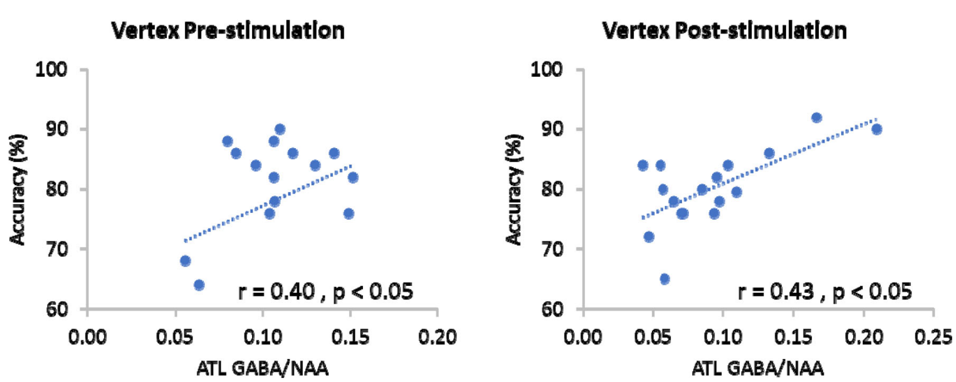

Figure S2.

The relationship between the ATL GABA levels and semantic task performance in the vertex stimulation.

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Reviewer #1 (Public Review):

Summary:
This study focuses on the role of GABA in semantic memory and its neuroplasticity. The researchers stimulated the left ATL and control site (vertex) using cTBS, measured changes in GABA before and after stimulation using MRS, and measured changes in BOLD signals during semantic and control tasks using fMRI. They analyzed the effects of stimulation on GABA, BOLD, and behavioral data, as well as the correlation between GABA changes and BOLD changes caused by the stimulation. The authors also analyzed the relationship between individual differences in GABA levels and behavioral performance in the semantic task. They found that cTBS stimulation led to increased GABA levels and decreased BOLD activity in the ATL, and these two changes were highly correlated. However, cTBS stimulation did not significantly change participants' behavioral performance on the semantic task, although behavioral changes in the control task were found after stimulation. Individual levels of GABA were significantly correlated with individuals' accuracy on the semantic task, and the inverted U-shaped (quadratic) function provides a better fit than the linear relationship. The authors argued that the results support the view that GABAergic inhibition can sharpen activated distributed semantic representations. They also claimed that the results revealed, for the first time, a non-linear, inverted-U-shape relationship between GABA levels in the ATL and semantic function, by explaining individual differences in semantic task performance and cTBS responsiveness

Strengths:
The findings of the research regarding the increase of GABA and decrease of BOLD caused by cTBS, as well as the correlation between the two, appear to be reliable. This should be valuable for understanding the biological effects of cTBS.

Weaknesses:
Regarding the behavioral effects of GABA on semantic tasks, especially its impact on neuroplasticity, the results presented in the article are inadequate to support the claims made by the authors. There are three aspects of results related to this: 1) the effects of cTBS stimulation on behavior, 2) the positive correlation between GABA levels and semantic task accuracy, and 3) the nonlinear relationship between GABA levels and semantic task accuracy. Among these three pieces of evidence, the clearest one is the positive correlation between GABA levels and semantic task accuracy. However, it is important to note that this correlation already exists before the stimulation, and there are no results supporting that it can be modulated by the stimulation. In fact, cTBS significantly increases GABA levels but does not significantly improve performance on semantic tasks. According to the authors' interpretation of the results in Table 1, cTBS stimulation may have masked the practice effects that were supposed to occur. In other words, the stimulation decreased rather than enhanced participants' behavioral performance on the semantic task.

The stimulation effect on behavioral performance could potentially be explained by the nonlinear relationship between GABA and performance on semantic tasks proposed by the authors. However, the current results are also insufficient to support the authors' hypothesis of an inverted U-shaped curve. Firstly, in Figure 3C and Figure 3D, the last one-third of the inverted U-shaped curve does not have any data points. In other words, as the GABA level increases the accuracy of the behavior first rises and then remains at a high level. This pattern of results may be due to the ceiling effect of the behavioral task's accuracy, rather than an inverted U-shaped ATL GABA function in semantic memory. Second, the article does not provide sufficient evidence to support the existence of an optimal level of GABA in the ATL. Fortunately, this can be tested with additional data analysis. The authors can estimate, based on pre-stimulus data from individuals, the optimal level of GABA for semantic functioning. They can then examine two expectations: first, participants with pre-stimulus GABA levels below the optimal level should show improved behavioral performance after stimulation-induced GABA elevation; second, participants with pre-stimulus GABA levels above the optimal level should exhibit a decline in behavioral performance after stimulation-induced GABA elevation. Alternatively, the authors can categorize participants into groups based on whether their behavioral performance improves or declines after stimulation, and compare the pre- and post-stimulus GABA levels between the two groups. If the improvement group shows significantly lower pre-stimulus GABA levels compared to the decline group, and both groups exhibit an increase in GABA levels after stimulation, this would also provide some support for the authors' hypothesis.

Another issue in this study is the confounding of simulation effects and practice effects. According to the results, there is a significant improvement in performance after the simulation, at least in the control task, which the authors suggest may reflect a practice effect. The authors argue that the results in Table 1 suggest a similar practice effect in the semantic task, but it is masked by the simulation of the ATL. However, since no significant effects were found in the ANOVA analysis of the semantic task, it is actually difficult to draw a conclusion. This potential confound increases the risk in data analysis and interpretation. Specifically, for Figure 3D, if practice effects are taken into account, the data before and after the simulation should not be analyzed together.

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Reviewer #2 (Public Review):

Summary:
The authors combined inhibitory neurostimulation (continuous theta-burst stimulation, cTBS) with subsequent MRI measurements to investigate the impact of inhibition of the left

anterior temporal lobe (ATL) on task-related activity and performance during a semantic task and link stimulation-induced changes to the neurochemical level by including MR spectroscopy (MRS). cTBS effects in the ATL were compared with a control site in the vertex. The authors found that relative to stimulation of the vertex, cTBS significantly increased the local GABA concentration in the ATL. cTBS also decreased task-related semantic activity in the ATL and potentially delayed semantic task performance by hindering a practice effect from pre to post. Finally, pooled data from their previous MRS study suggest an inverted U-shape between GABA concentration and behavioral performance. These results help to better understand the neuromodulatory effects of non-invasive brain stimulation on task performance.

Strengths:
Multimodal assessment of neurostimulation effects on the behavioral, neurochemical, and neural levels. In particular, the link between GABA modulation and behavior is timely and potentially interesting.

Weaknesses:
The analyses are not sound. Some of the effects are very weak and not all conclusions are supported by the data since some of the comparisons are not justified. There is some redundancy with a previous paper by the same authors, so the novelty and contribution to the field are overall limited. A network approach might help here.

https://doi.org/10.7554/eLife.91771.1.sa1

Reviewer #3 (Public Review):

Summary:
The authors used cTBS TMS, magnetic resonance spectroscopy (MRS), and functional magnetic resonance imaging (fMRI) as the main methods of investigation. Their data show that cTBS modulates GABA concentration and task-dependent BOLD in the ATL, whereby greater GABA increase following ATL cTBS showed greater reductions in BOLD changes in ATL. This effect was also reflected in the performance of the behavioural task response times, which did not subsume to practice effects after AL cTBS as opposed to the associated control site and control task. This is in line with their first hypothesis. The data further indicates that regional GABA concentrations in the ATL play a crucial role in semantic memory because individuals with higher (but not excessive) GABA concentrations in the ATLs performed better on the semantic task. This is in line with their second prediction. Finally, the authors conducted additional analyses to explore the mechanistic link between ATL inhibitory GABAergic action and semantic task performance. They show that this link is best captured by an inverted U-shaped function as a result of a quadratic linear regression model. Fitting this model to their data indicates that increasing GABA levels led to better task performance as long as they were not excessively low or excessively high. This was first tested as a relationship between GABA levels in the ATL and semantic task performance; then the same analyses were performed on the pre and post-cTBS TMS stimulation data, showing the same pattern. These results are in line with the conclusions of the authors.

Strengths:
I thoroughly enjoyed reading the manuscript and appreciate its contribution to the field of the role of the ATL in semantic processing, especially given the efforts to overcome the immense challenges of investigating ATL function by neuroscientific methods such as MRS, fMRI & TMS. The main strengths are summarised as follows:

• The work is methodologically rigorous and dwells on complex and complementary multimethod approaches implemented to inform about ATL function in semantic memory as reflected in changes in regional GABA concentrations. Although the authors previously demonstrated a negative relationship between increased GABA levels and BOLD signal changes during semantic processing, the unique contribution of this work lies within evidence on the effects of cTBS TMS over the ATL given by direct observations of GABA concentration changes and further exploring inter-individual variability in ATL neuroplasticity and consequent semantic task performance.

• Another major asset of the present study is implementing a quadratic regression model to provide insights into the non-linear relationship between inhibitory GABAergic activity within the ATLs and semantic cognition, which improves with increasing GABA levels but only as long as GABA levels are not extremely high or low. Based on this finding, the authors further pinpoint the role of inter-individual differences in GABA levels and cTBS TMS responsiveness, which is a novel explanation not previously considered (according to my best knowledge) in research investigating the effect of TMS on ATLs.

• There are also many examples of good research practice throughout the manuscript, such as the explicitly stated exploratory analyses, calculation of TMS electric fields, using ATL optimised dual echo fRMI, links to open source resources, and a part of data replicates a previous study by Jung et. al (2017).

Weaknesses:

• Research on the role of neurotransmitters in semantic memory is still very rare and therefore the manuscript would benefit from more context on how GABA contributes to individual differences in cognition/behaviour and more justification on why the focus is on semantic memory. A recommendation to the authors is to highlight and explain in more depth the particular gaps in evidence in this regard.

• The focus across the experiments is on the left ATL; how do the authors justify this decision? Highlighting the justification for this methodological decision will be important, especially given that a substantial body of evidence suggests that the ATL should be involved in semantics bilaterally (e.g. Hoffman & Lambon Ralph, 2018; Lambon Ralph et al., 2009; Rice et al., 2017; Rice, Hoffman, et al., 2015; Rice, Ralph, et al., 2015; Visser et al., 2010).

• When describing the results, (Pg. 11; lines 233-243), the authors first show that the higher the BOLD signal intensity in ATL as a response to the semantic task, the lower the GABA concentration. Then, they state that individuals with higher GABA concentrations in the ATL perform the semantic task better. Although it becomes clearer with the exploratory analysis described later, at this point, the results seem rather contradictory and make the reader question the following: if increased GABA leads to less task-induced ATL activation, why at this point increased GABA also leads to facilitating and not inhibiting semantic task performance? It would be beneficial to acknowledge this contradiction and explain how the following analyses will address this discrepancy.

• There is an inconsistency in reporting behavioural outcomes from the performance on the semantic task. While experiment 1 (cTBS modulates regional GANA concentrations and task-related BOLD signal changes in the ATL) reports the effects of cTBS TMS on response times, experiment 2 (Regional GABA concentrations in the ATL play a crucial role in semantic memory) and experiment 3 (The inverted U-shaped function of ATL GABA concentration in semantic processing) report results on accuracy. For full transparency, the manuscript would benefit from reporting all results (either in the main text or supplementary materials) and providing further explanations on why only one or the other outcome is sensitive to the experimental manipulations across the three experiments.

Overall, the most notable impact of this work is the contribution to a better understanding of individual differences in semantic behaviour and the potential to guide therapeutic interventions to restore semantic abilities in neurological populations. While I appreciate that this is certainly the case, I would be curious to read more about how this could be achieved.

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